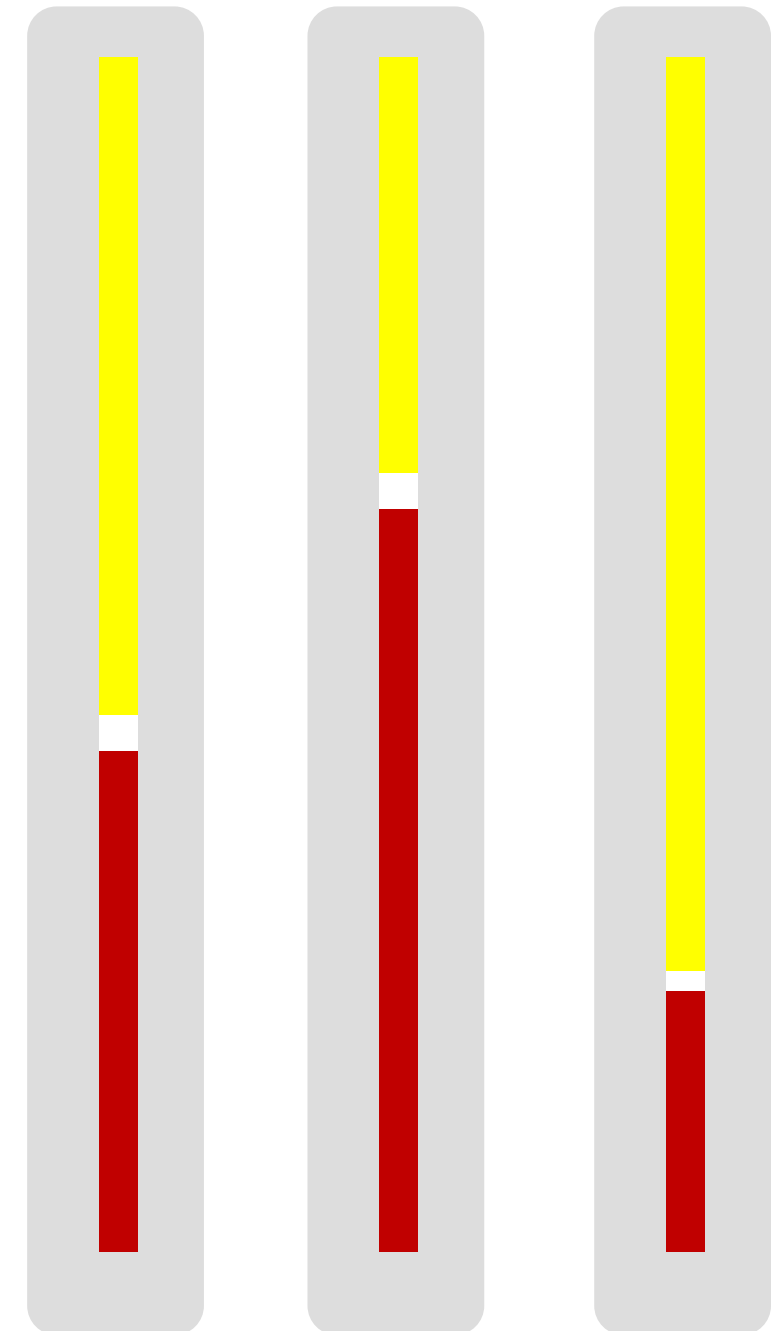


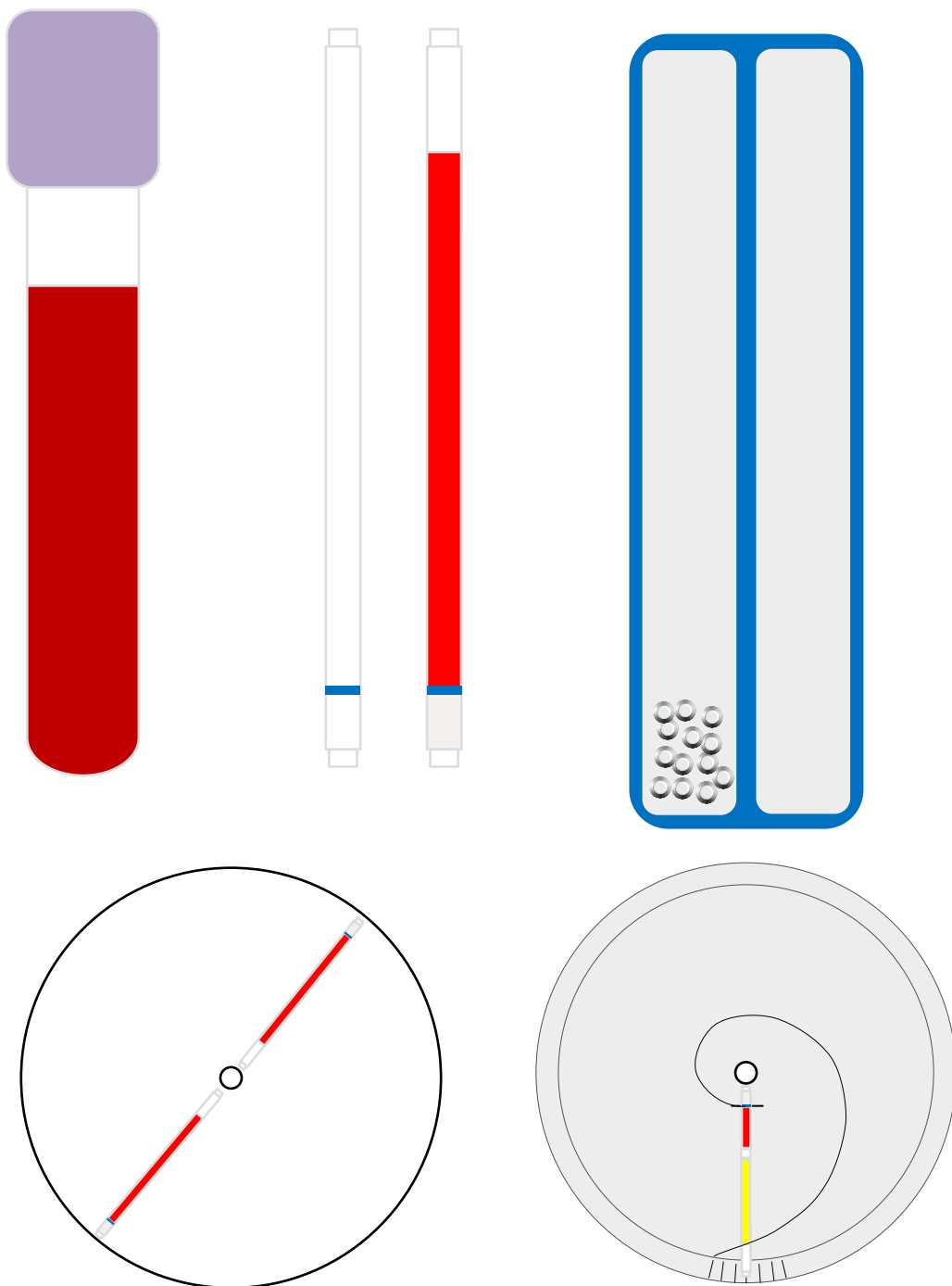
Spun Hematocrit

Principle

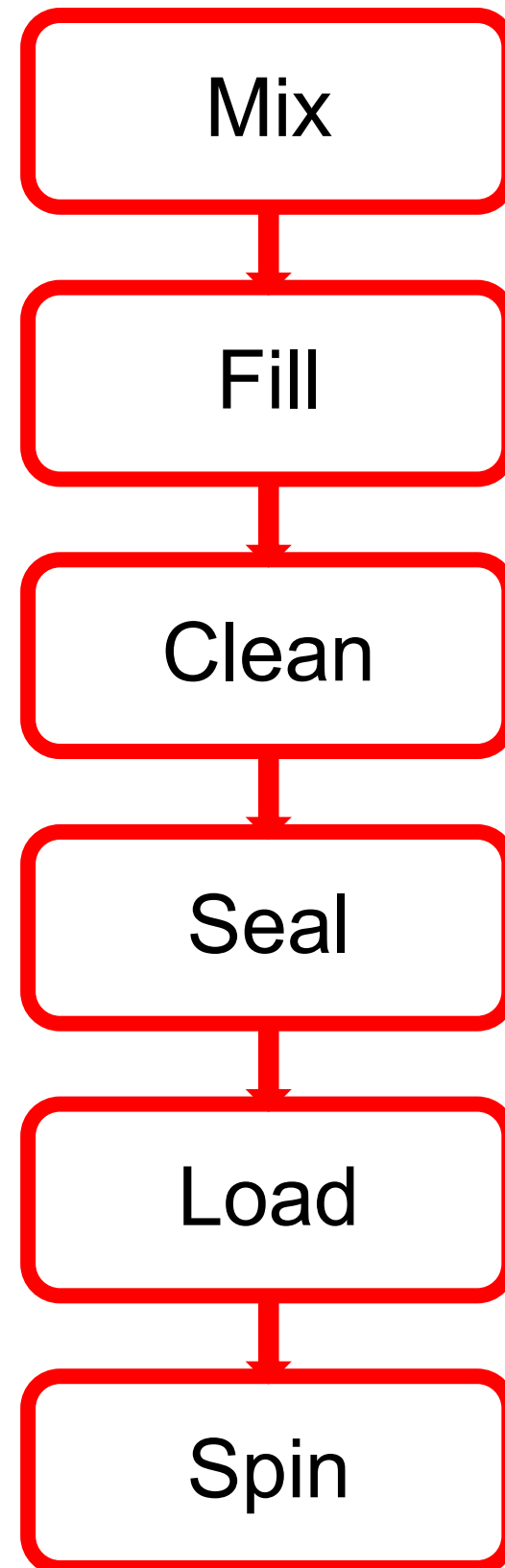
- **Definition:** Hematocrit (HCT), also known as Packed Cell Volume (PCV)
 - Ratio of the volume of erythrocytes (RBCs) to the total volume of whole blood
- **Expression:** Percentage (%) or a decimal fraction (L/L).
- **Mechanism:**
 1. Whole blood in a capillary tube is centrifuged (~ 5,000 rpm for 5 minutes)
 2. Centrifugal force separates whole blood into three distinct layers:
 - **Top Layer:** Clear Plasma
 - **Middle Layer:** Buffy Coat (WBCs & Platelets)
 - **Bottom Layer:** Packed Red Blood Cells (RBCs)
 3. Packed RBC column is measured relative to total blood column length



Equipment & Supplies



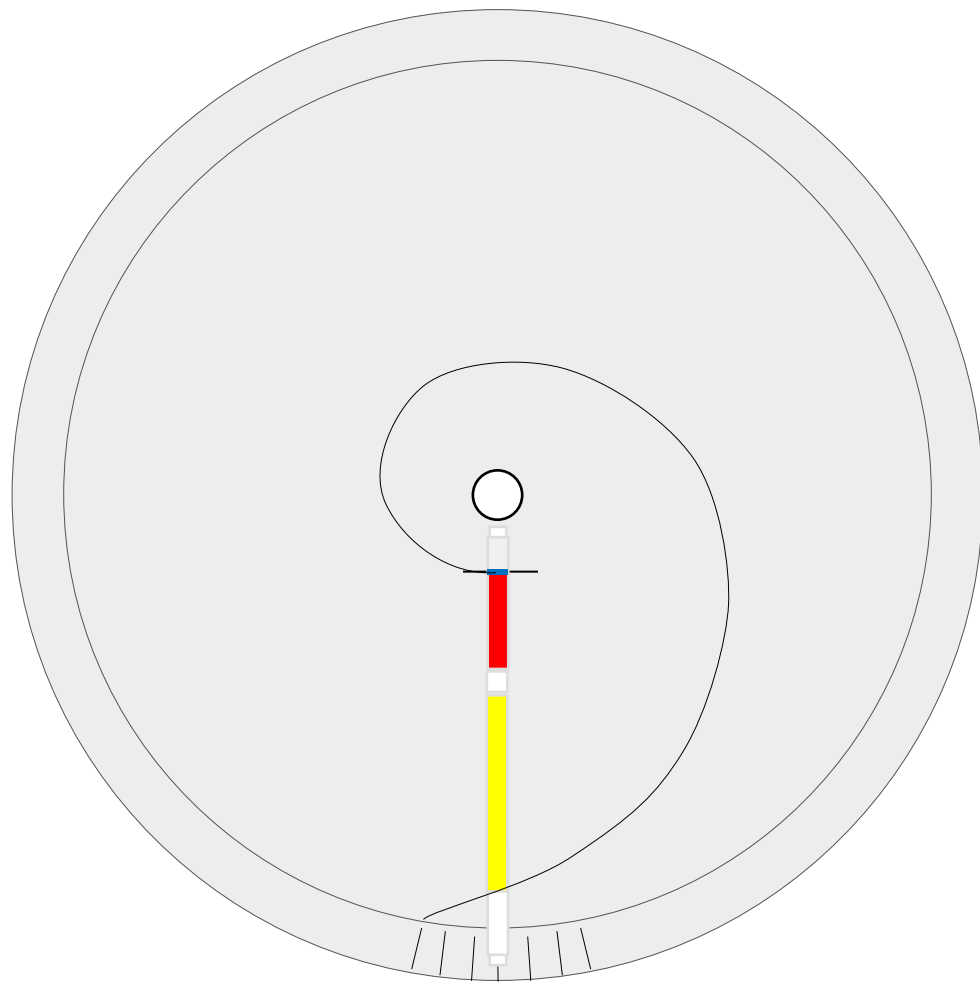
- **Specimen:** EDTA whole blood
 - Must be processed within 4 hours of collection
 - Tube must be free of micro-clots
- **Capillary Tubes:** Plain capillary tubes of constant internal bore (1.1–1.2mm)
- **Sealing Compound:** Critoseal (clay plug)
- **Centrifuge:** Microhematocrit centrifuge
- **Reading Device:** Microhematocrit Reader disk
- **Gauze/Wipes**



Procedure

1. **Mixing:** Thoroughly mix the EDTA blood tube by gentle inversion
2. **Filling Tubes:** Fill two capillary tubes 2/3 to 3/4 full by capillary action
3. **Cleaning:** Carefully wipe the outside of capillary tubes with clean gauze.
4. **Sealing:** Plug one end of each tube with Critoseal clay
5. **Centrifuge Loading:** Place tubes into microhematocrit centrifuge slots:
 - **CRITICAL:** Sealed clay ends **MUST** face outward
 - Record capillary tube positions
6. **Centrifugation:** Spin at ~5,000 rpm for ~5 minutes

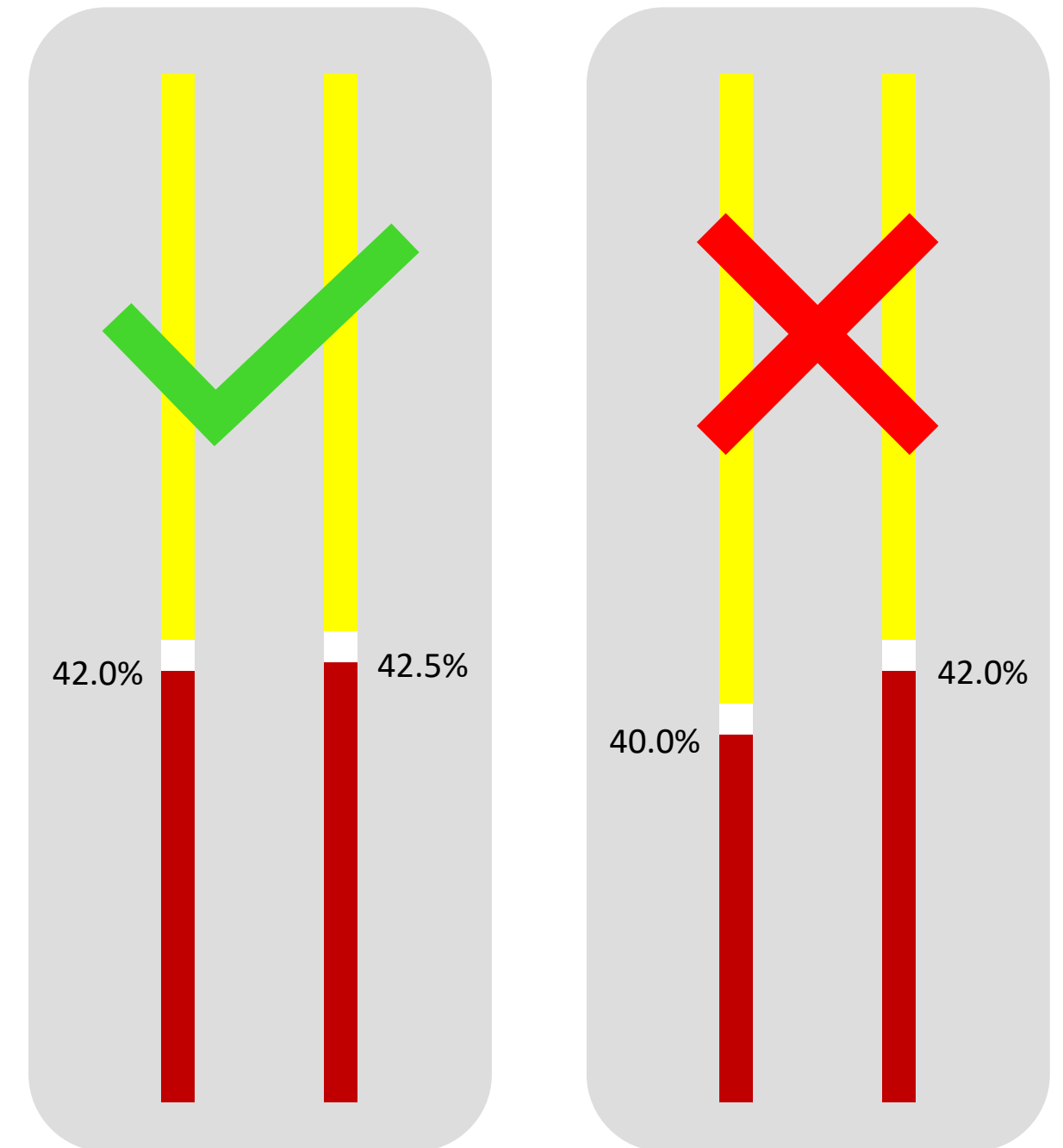
Microhematocrit Reader



1. **Positioning:** Place capillary tube on the reader base disk with the sealed clay end pointing toward the center
2. **Clay Alignment:** Adjust tube so the line separating the Critoseal clay and RBCs aligns exactly with the bottom black line (0% mark)
3. **Plasma Alignment:** Rotate the entire reader assembly so the outer pin aligns with the 100% mark, then hold the bottom disk stationary and rotate the upper disk until the curved line touches the top of the plasma column
4. **HCT Measurement:** Rotate the reader disk until the curved line intersects the junction of the RBC column and buffy coat (excluding the buffy coat).
5. **Recording:** Read the value on the percentage scale (read to the nearest 0.5%)

Duplicate Precision & Quality Control

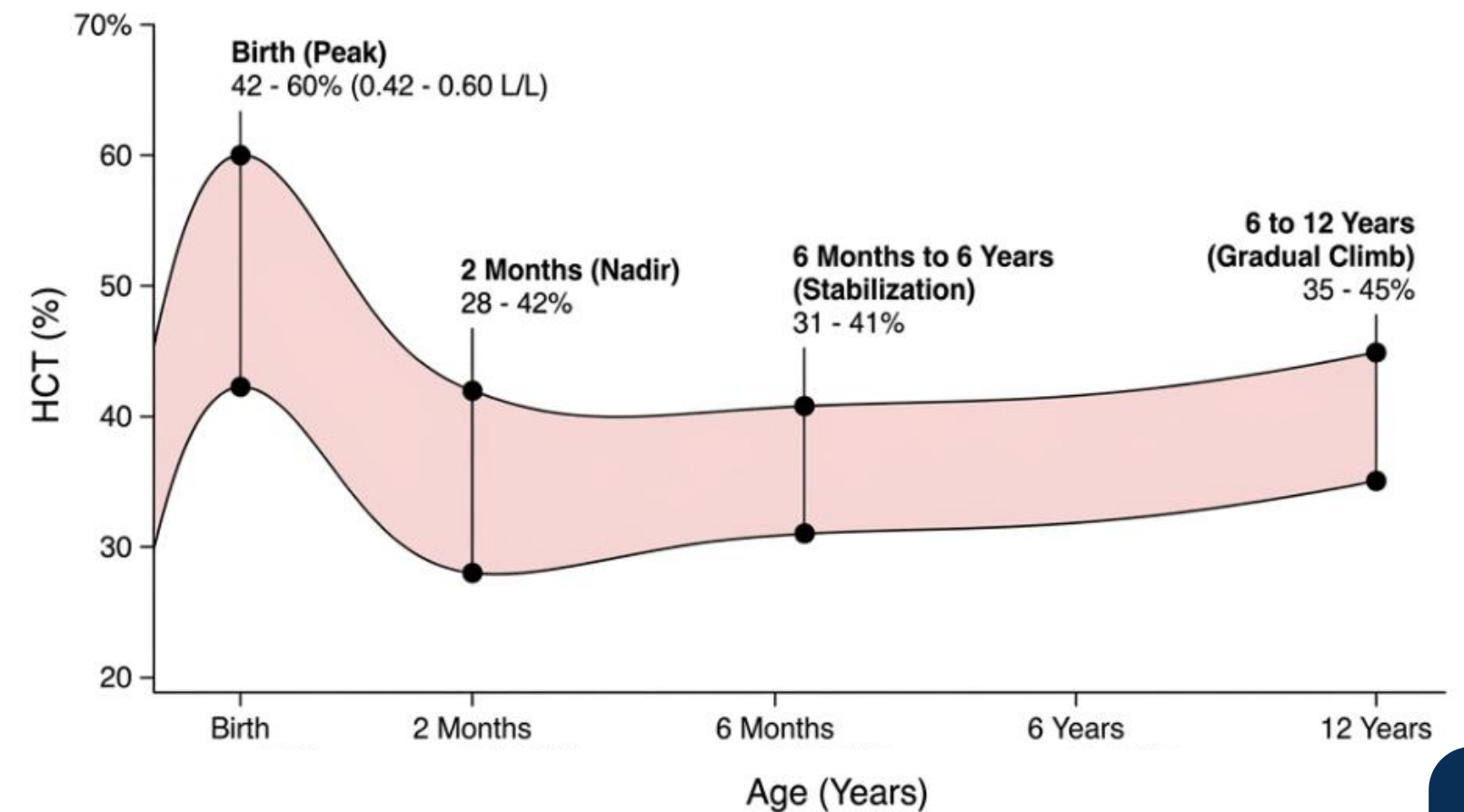
- **Duplicate Testing Required:** Manual hematocrits must be run in duplicate (two capillary tubes per patient sample)
- **Acceptance Limit:** Duplicate results must agree within $\pm 1.0\%$ (or 1 percentage point)
 - *Example:* Tube 1 = 42.0%, Tube 2 = 42.5%
 - $\Rightarrow$ ACCEPTABLE (Difference = 0.5% and $< 1\%$)
 - *Example:* Tube 1 = 40.0%, Tube 2 = 42.0%
 - $\Rightarrow$ REJECT (Difference = 2.0% and $> 1\%$)
- **Reporting:** Report final HCT as the average of the duplicate tubes rounded to the nearest 0.1%
- **Control Out-of-Range:** Do not report patient results if commercial controls are out of range



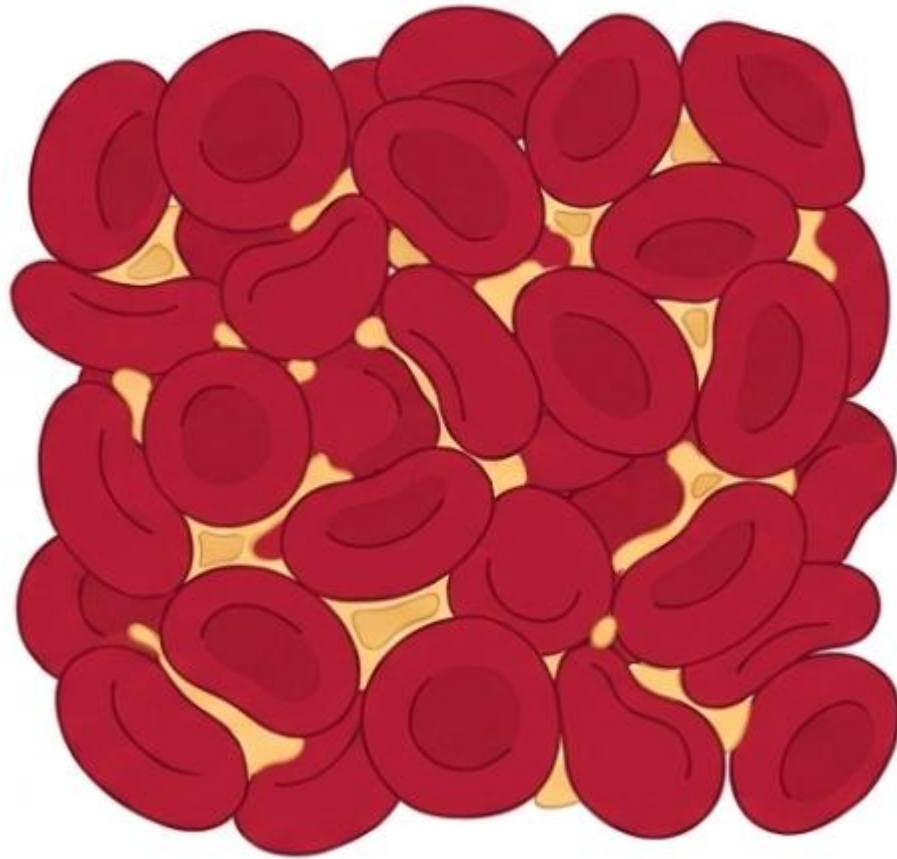
Reference Intervals

Age Group	Reference Interval (%)	Reference Interval (L/L)
Newborn (Birth)	42 – 60%	0.42 – 0.60
Infant (2 weeks)	39 – 65%	0.39 – 0.65
Infant (1 month)	33 – 55%	0.33 – 0.55
Infant (2 months)	28 – 42%	0.28 – 0.42
Infant (6 months)	31 – 41%	0.31 – 0.41
Child (1 year)	33 – 41%	0.33 – 0.41
Child (2 - 6 years)	34 – 41%	0.34 – 0.41
Child (6 - 12 years)	35 – 45%	0.35 – 0.45

Population	Conventional Units (%)	SI Units (L/L)
Adult Males	42 – 52%	0.42 – 0.52 L/L
Adult Females	36 – 46%	0.36 – 0.46 L/L



Sources of Error & Clinical Artifacts



- **Trapped Plasma (+2% to 3% elevation):**
 - Small amounts of plasma remain trapped between packed RBCs
 - Spun HCT values are typically automated instrument values
 - Occurs in sickle cell anemia, spherocytosis, hypochromic anemias, and macrocytosis
- **Incomplete Tube Sealing:**
 - Clay plug fails $\Rightarrow$ RBCs and plasma leak during spin $\Rightarrow$ Falsely decreased HCT
- **Under-Centrifugation:**
 - Centrifuging for < 5 minutes or at low speed $\Rightarrow$ Incomplete packing $\Rightarrow$ Falsely elevated HCT

Technical Pitfalls & Safety Precautions

- **Buffy Coat Inclusion:**
 - Incorrectly including the white buffy coat layer in the RBC reading $\Rightarrow$ Falsely elevated HCT
- **Specimen Hemolysis / Clots:**
 - Clotted blood $\Rightarrow$ Falsely decreased HCT
 - Hemolyzed sample $\Rightarrow$ Falsely decreased HCT
- **CRITICAL SAFETY WARNING:**
 - **NEVER** attempt to stop the microhematocrit centrifuge by applying hand pressure to the spinning rotor or lid plate!
 - **Consequences:** Causes the RBC column to "sling" forward (falsely elevated HCT) and risks severe mechanical injury or tube breakage